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Investigation of the effect of different types of materials on the shielding of alpha particles emitted by Americium-241

Graphic from Canva

ABSTRACT:

- This research paper explores how different materials shield alpha particles from Americium-241.
- We hypothesised that most materials can block alpha radiation. A sheet of material and americium-241 was placed in front of a Geiger counter for 3 minutes, and counts were averaged per minute.
- Results showed Carbon Fibre provided the best shielding, followed by Baking Paper. Paper had the least shielding,
- Instead of rubber, leather, or cloth, Thick Carbon Fibre with Baking Paper elements can improve radiation protection, offering a cost-effective alternative to hazmat suits.
- This supports the UNSDG of good health & wellbeing by reducing radiation-related injuries.

QUESTION/PROBLEM:

What is the relationship between different types of materials on the shielding of alpha particles released by Americium-241?

PROCEDURE

- PLACE THE GEIGER COUNTER AND PVC PLASTIC SHEET ON RETORT STANDS; THE GEIGER COUNTER AND AMERICIUM ARE 2 CM AWAY FROM THE PVC PLASTIC SHEET
- WAIT 3 MINUTES FOR THE RADIATION TO PASS THROUGH
- RECORD TOTAL COUNT
- REPEAT STEPS 1-3 FOR EVERY MATERIAL, WHERE THE MATERIAL IS USED AS THE INDEPENDENT VARIABLE.
- COMPARE THE AVERAGE AMOUNT OF RADIATION COUNTED (IN CPM) AFTER PASSING THROUGH THE 5 MATERIALS.

Graphic from Canva

RESULTS

Material	Total Counts in 3 min	Avg Count in 1 min
Paper	311.00	103.67
Cardstock	297.00	99.00
Aluminium foil	309.00	103.00
PVC Plastic	290.00	96.67
Vinyl Sheet	302	100.67
Baking paper	283	94.33
No Material	330.00	110.00
Carbon Fiber	272.00	90.66

Diagram created by Lee Heyi

OUR RESULTS

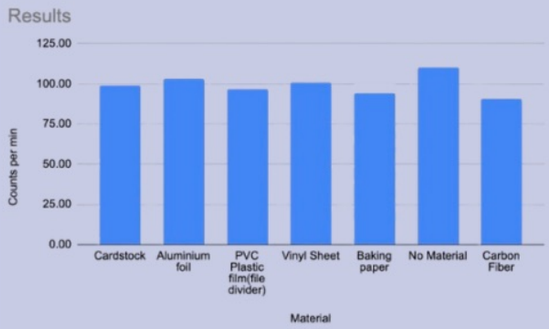
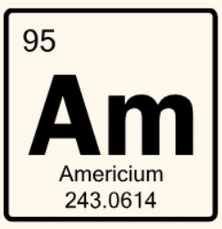


Diagram created by Avyan Mehra

HYPOTHESIS:

We hypothesise that most types of materials tested will be able to block most of the alpha particles released by the Americium.



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HYPOTHESIS

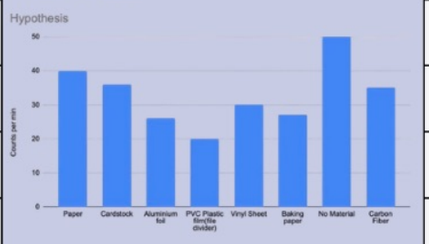

Diagram created by Avyan Mehra

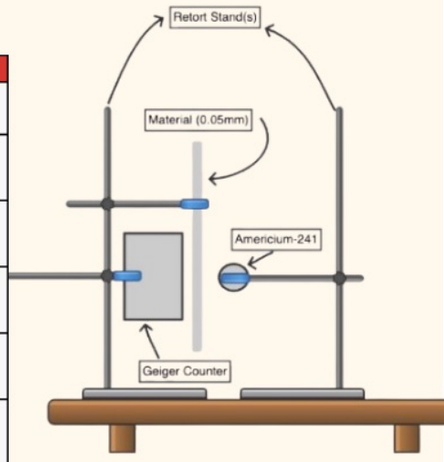

Diagram created by Avyan Mehra

CONCLUSION

- Aluminum foil and Carbon fiber provided moderate shielding
- Lighter materials like paper and cardstock were far less effective.
- Surprisingly, baking paper and vinyl sheets also offered limited protection.
- We initially hypothesised that most materials would block a substantial amount of radiation, but our results disproved this.

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DISCUSSION ON RESULTS

- Carbon Fibre is the best shield of those that we tested.
- Other materials such as baking paper, PVC plastic film, cardstock and vinyl sheets were able to shield some radiation.
- However, the difference is very slight. The difference between the amount of radiation shielded by the Carbon Fibre and the baking paper is 3.67%
- Both plastic products were able to shield against the background radiation relatively well compared to the other materials, such as Paper and Baking Paper.(Which let 115 and 112 counts through, respectively)
- It has been proven that thick pieces of plastic have the ability to block Alpha particles. (Unknown, 2023) This is because they are less dense and can stop the particles from piercing through(Canada Metal, 2021)
- This is because high-energy beta particles must be shielded by materials with a low atomic weight. This includes plastic. (Perkins, S., 2024, February 21)
- Another research paper (Cameron Dargahi, Bradley Lloyd, Alexis Deukam, 2015) shows that while plastic effectively shields against radiation, it is unreliable in high-temperature environments.